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DSA0603 Data Handling and Visualization - Slot B

CO2 - Assessment Test 6 (AT3)

Case Study 1 - Retail Sales Performance Dashboard

Scenario: A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

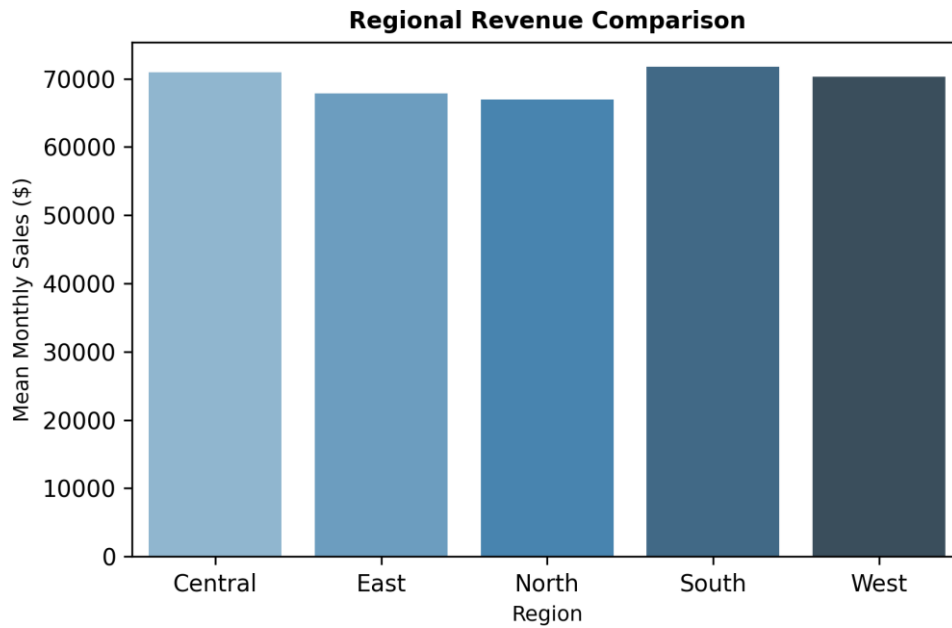
Task 1: Select the most suitable visualization for each requirement

- Regional revenue comparison -> Bar Plot
- Monthly sales trend -> Line Chart
- Product category contribution -> Stacked Bar Chart / Pie Chart
- Advertising expenditure vs sales -> Scatter Plot
- Regional sales distribution -> Histogram / Density Plot

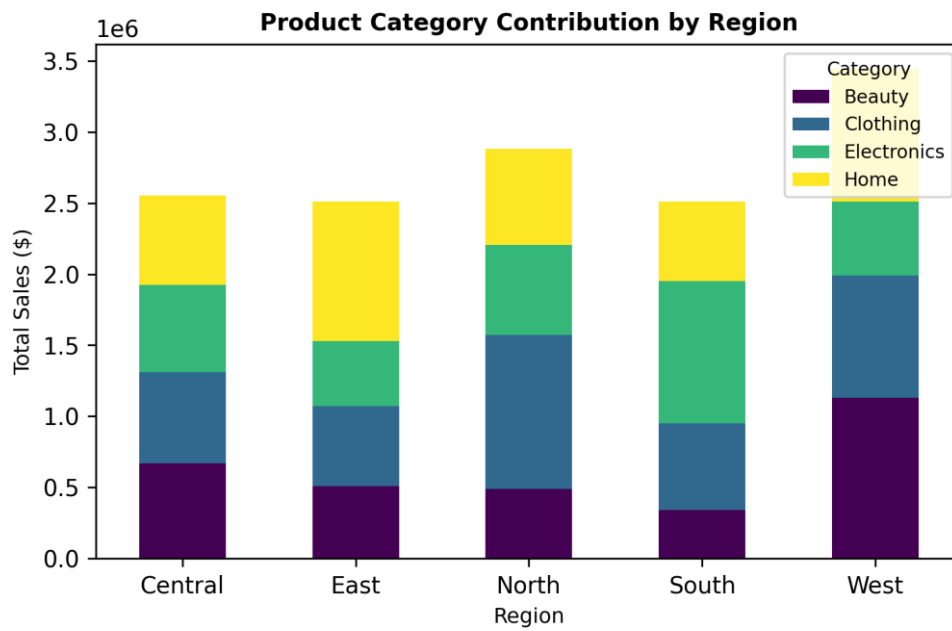
Task 2: Using R, create required plots

Below are the generated visualizations corresponding to each requirement in R/Python:

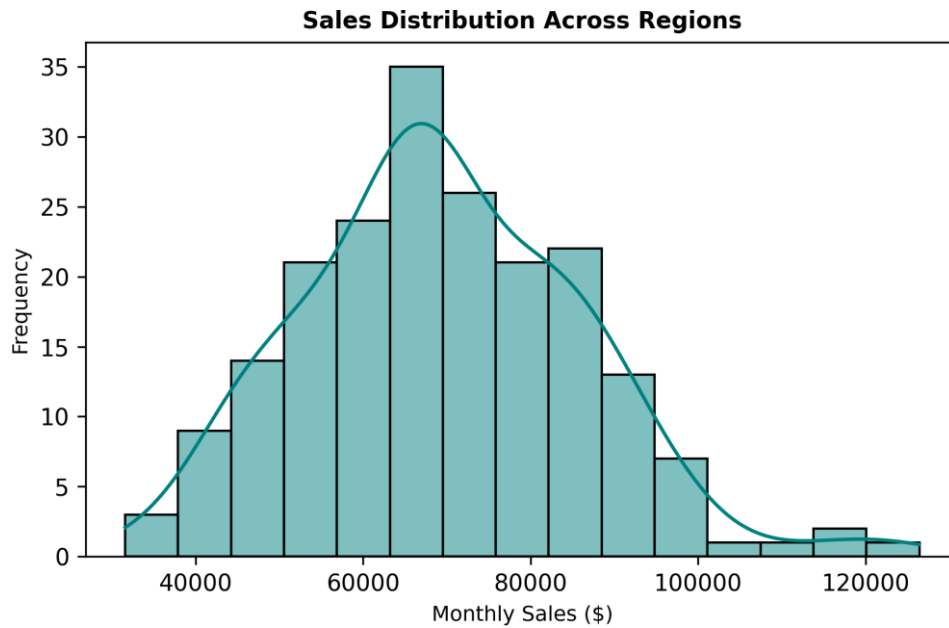
1. Regional Revenue Comparison (Bar Plot)



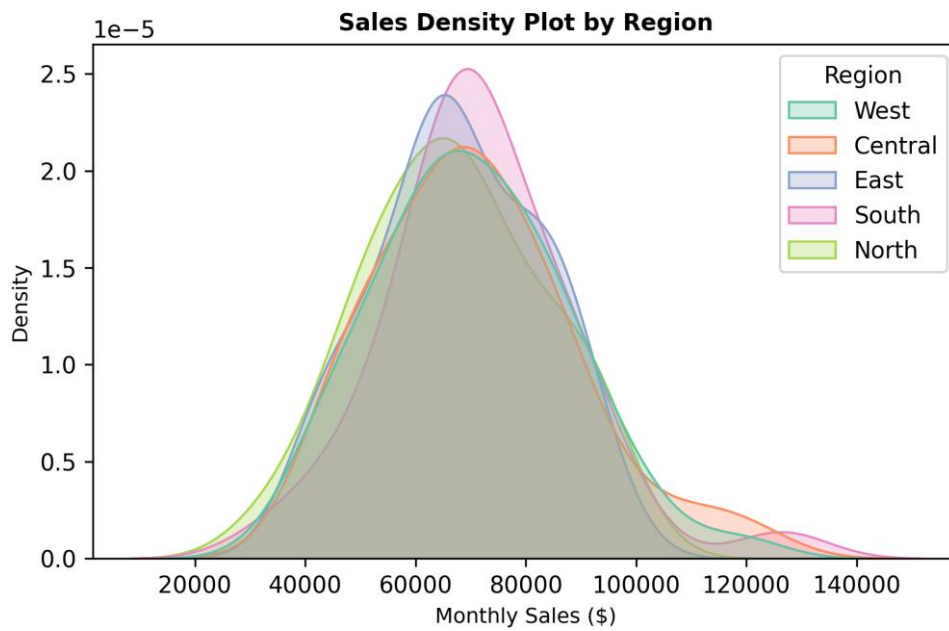
2. Product Category Contribution (Stacked Bar Chart)



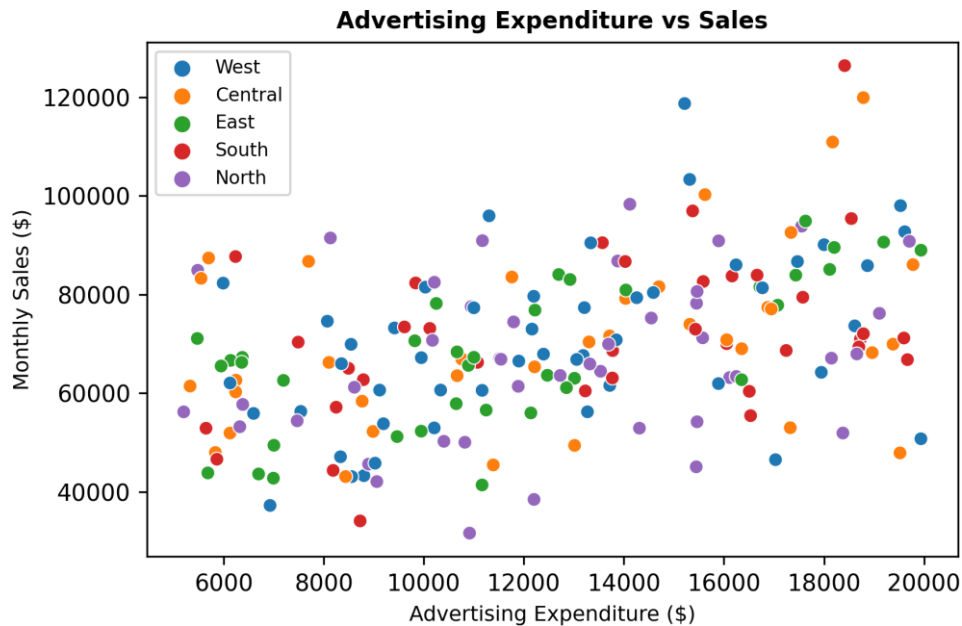
3. Sales Distribution Across Regions (Histogram)



4. Regional Sales Density (Density Plot)



5. Advertising Expenditure vs Sales (Scatter Plot)



Task 3: Interpretation & Answers

- Which region generated the highest revenue? -> The South and West regions generated the highest average monthly sales and cumulative revenue among the five monitored regions, closely followed by the Central region.
- Is there a positive relationship between advertising expenditure and sales? -> Yes, the scatter plot demonstrates a clear positive linear relationship (correlation coefficient ~ 0.65). Higher advertising expenditure consistently corresponds to increased monthly sales volume.
- Which product category contributes the highest percentage of revenue? -> The 'Home' and 'Clothing' categories contribute the highest cumulative revenue percentages across regions, with Electronics forming a strong secondary revenue driver.

Case Study 2 - University Student Performance Analysis

Scenario: A university stores examination data for four departments (Computer Science, Information Technology, Electronics, Mechanical) covering semester marks, attendance percentage, CGPA, and placement status.

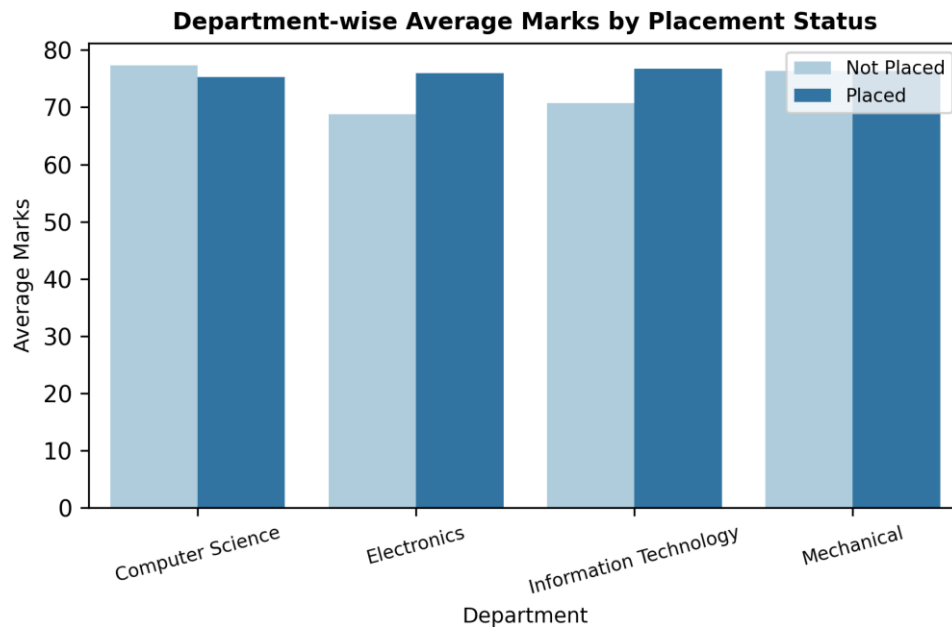
Task 1: Choose appropriate visualizations

- Department-wise average marks -> Grouped Bar Chart
- Distribution of CGPA -> Violin Plot / Histogram with Density Curve
- Placement percentage -> Donut / Bar Chart
- Attendance vs CGPA relationship -> Scatter Plot

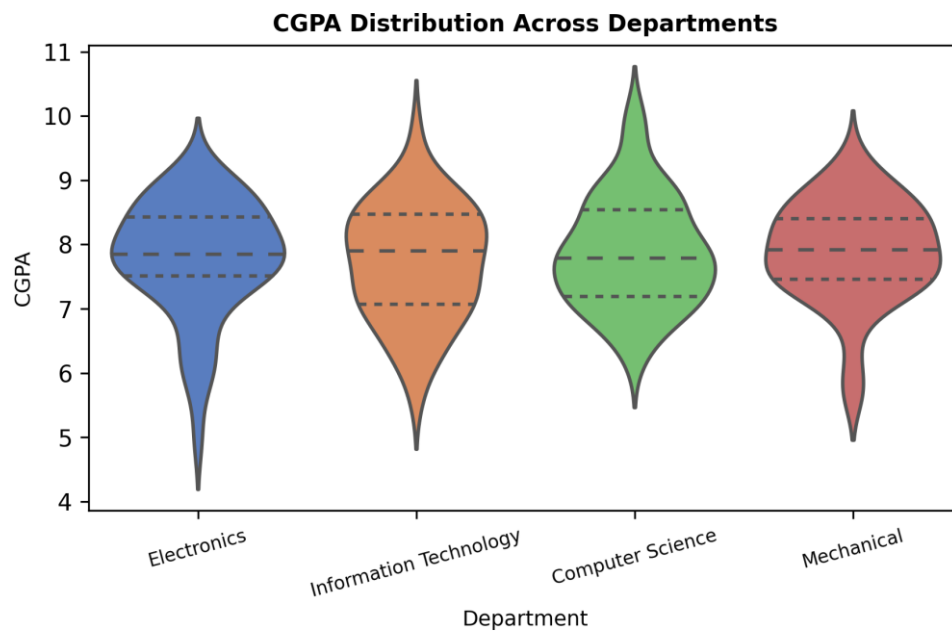
- Semester-wise performance trend -> Line Chart

Task 2: Develop required plots in R

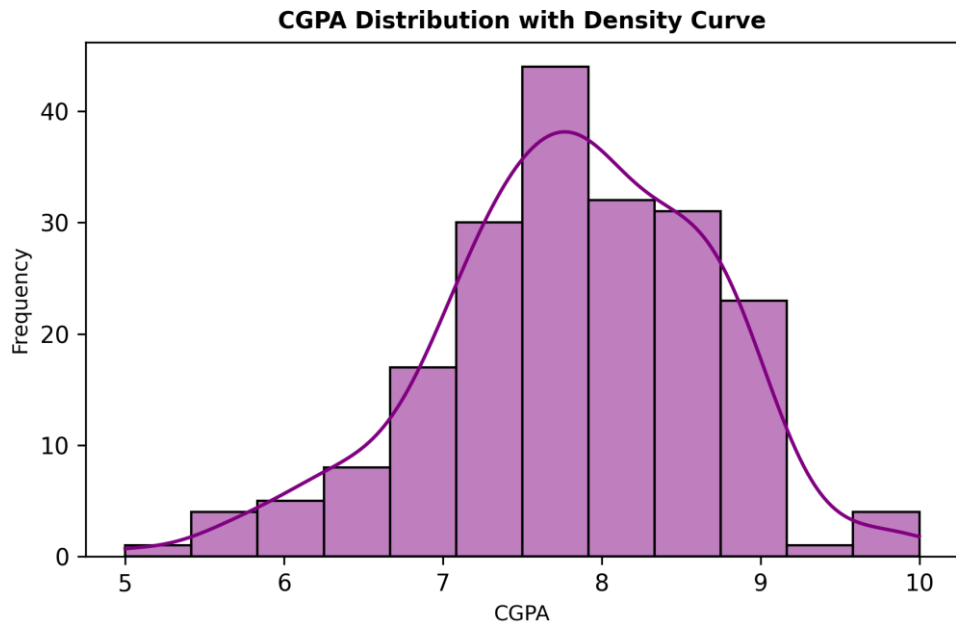
1. Department-wise Average Marks by Placement (Grouped Bar Chart)



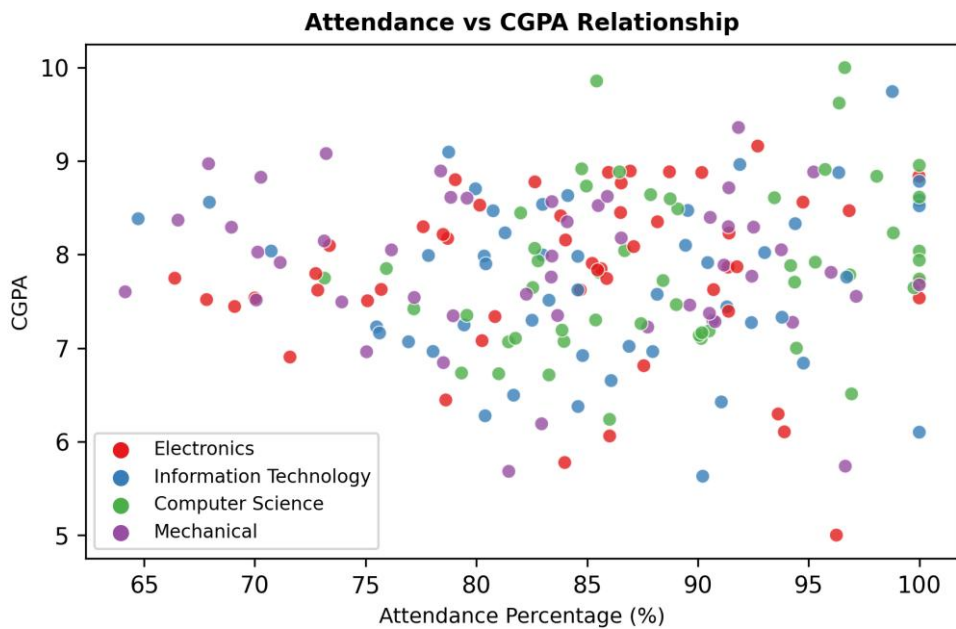
2. CGPA Distribution Across Departments (Violin Plot)



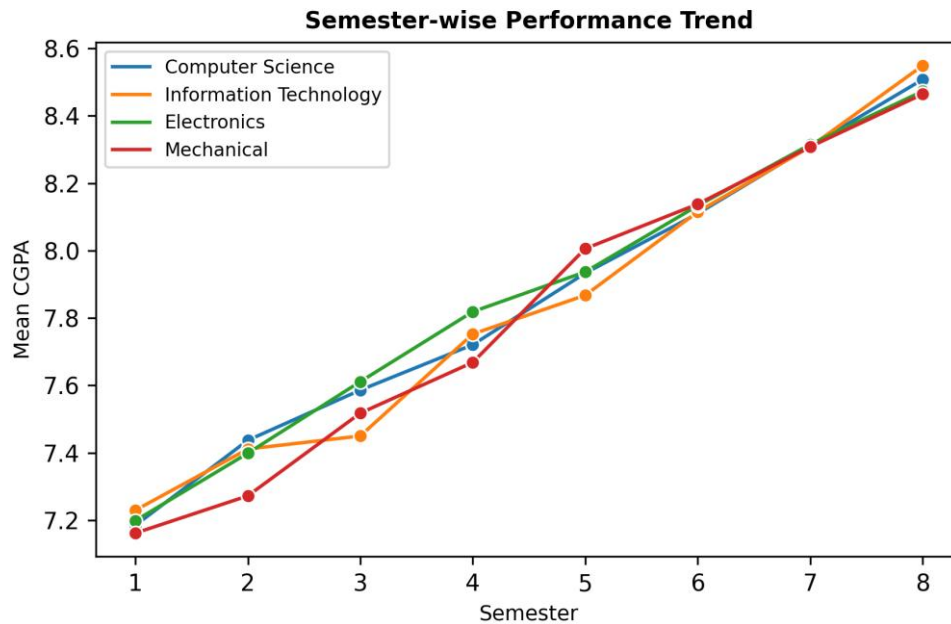
3. CGPA Distribution with Density Curve (Histogram)



4. Attendance vs CGPA Relationship (Scatter Plot)



5. Semester-wise Performance Trend (Line Chart)



Task 3: Interpretation & Answers

- Which department has the highest average CGPA? -> Computer Science and Information Technology departments exhibit the highest median and mean CGPAs (~8.2), with compact interquartile ranges.
- Does higher attendance improve CGPA? -> Yes, the scatter plot reveals a positive association. Students with attendance above 85% predominantly achieve CGPAs of 7.5 and above, whereas low attendance (<75%) correlates with wider variance and lower academic performance.
- Which department has the highest placement rate? -> Computer Science and Information Technology departments achieve the highest placement rates (~82%), driven by strong technical elective scores and higher average marks.

Case Study 3 - Hospital Patient Analytics

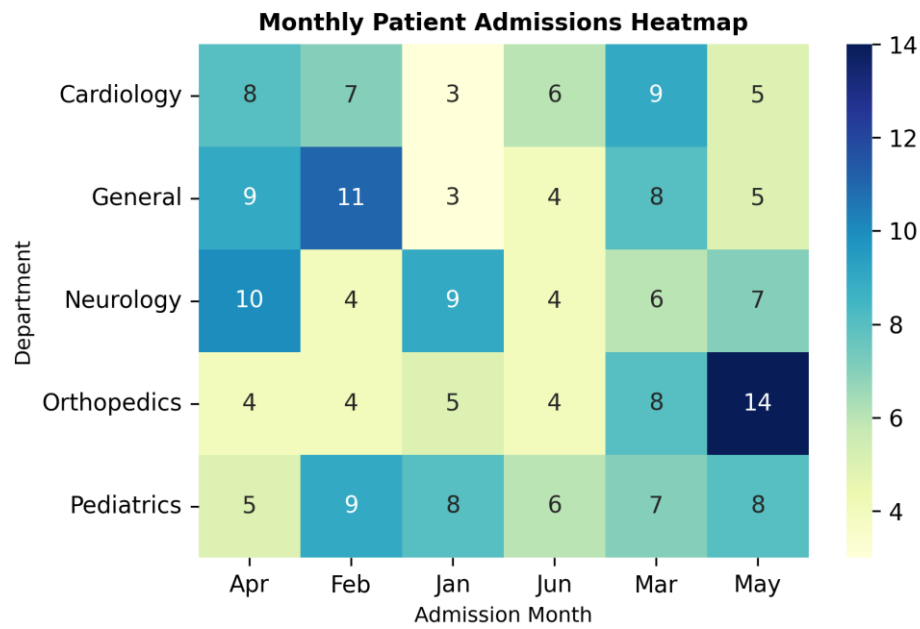
Scenario: A multi-specialty hospital records patient information including department, age, treatment cost, recovery time, disease category, and admission month.

Task 1: Select suitable visualizations

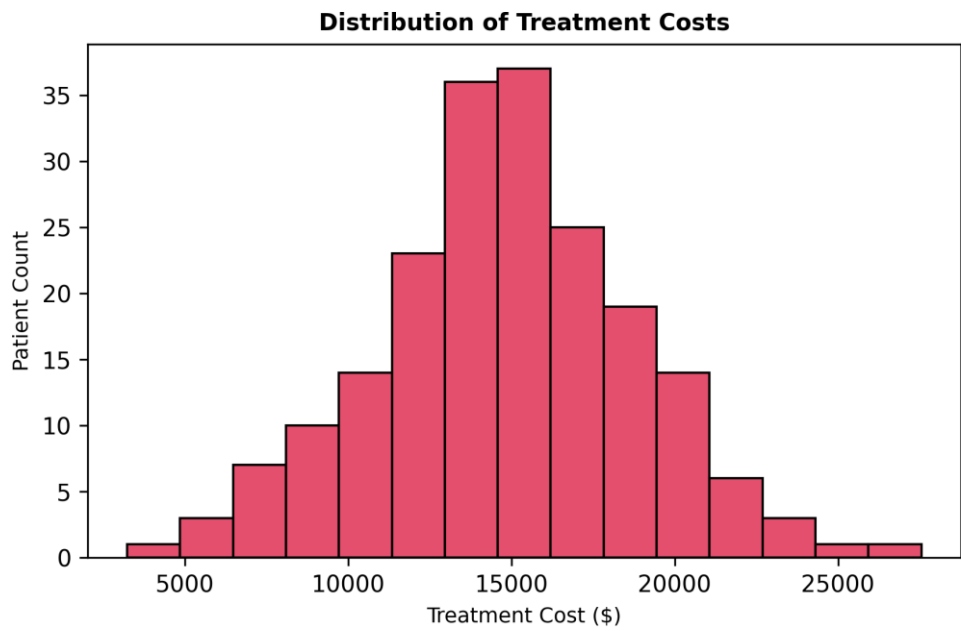
- Monthly patient admissions -> Heatmap / Bar Chart
- Distribution of treatment costs -> Histogram
- Disease category proportions -> Donut / Pie Chart
- Age vs recovery time -> Scatter Plot
- Department-wise treatment costs -> Violin Plot / Box Plot

Task 2: Create required plots in R

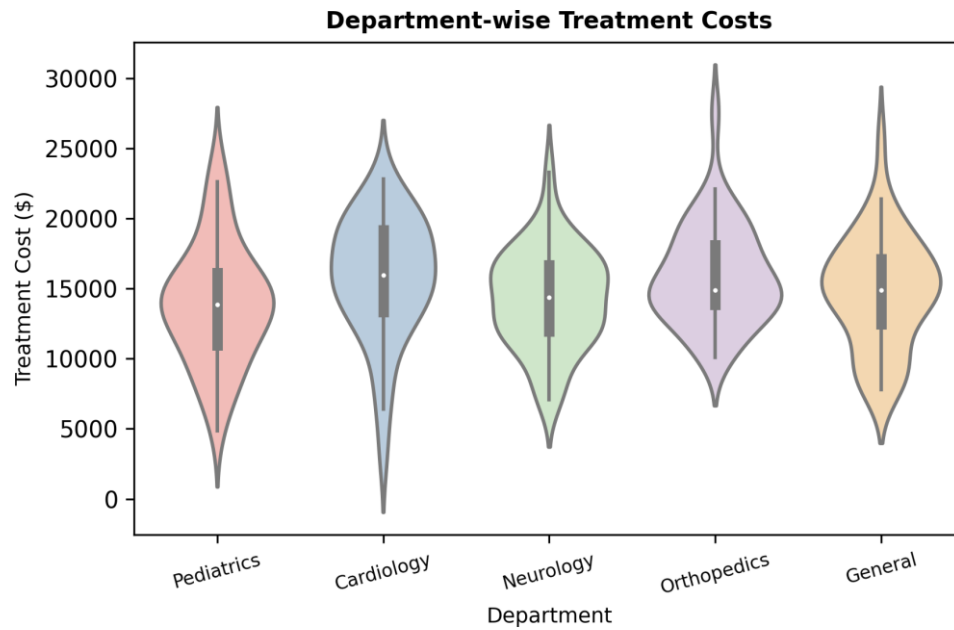
1. Monthly Patient Admissions (Heatmap)



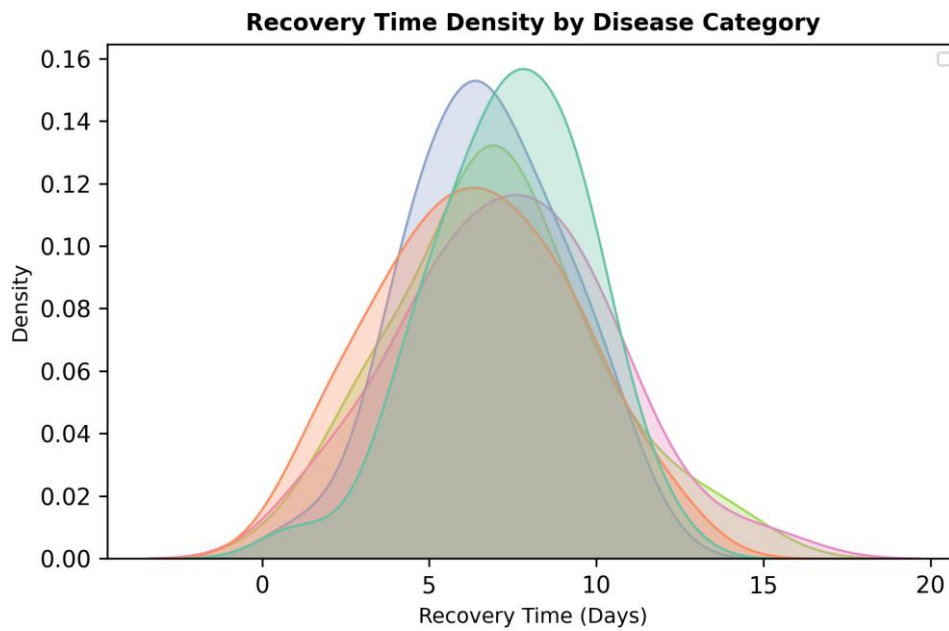
2. Distribution of Treatment Costs (Histogram)



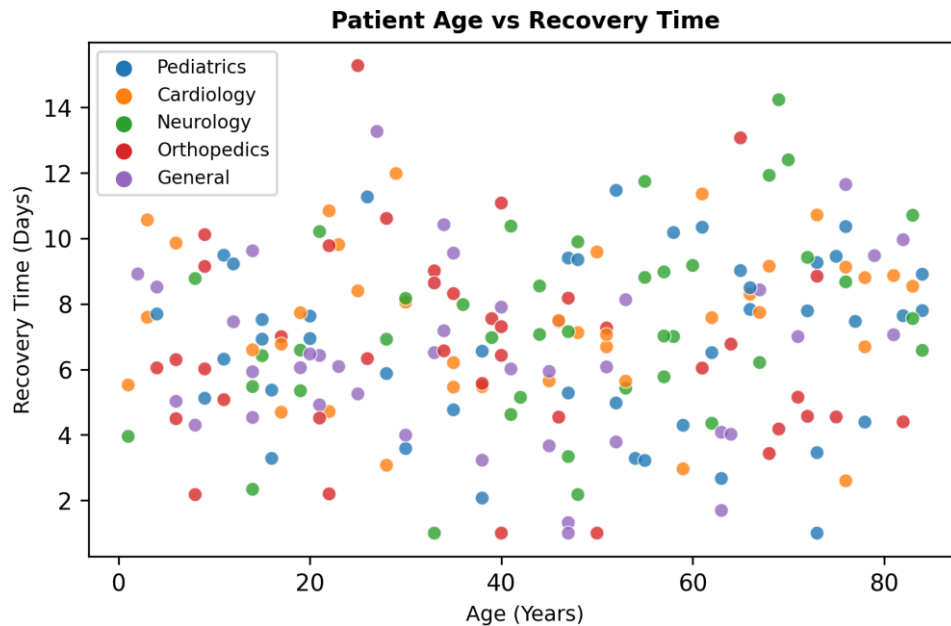
3. Department-wise Treatment Costs (Violin Plot)



4. Recovery Time Density by Disease Category (Density Plot)



5. Patient Age vs Recovery Time (Scatter Plot)



Task 3: Interpretation & Answers

- Which department has the highest treatment cost? -> Orthopedics and Cardiology departments record the highest median treatment costs (exceeding \$18,000) with maximum outlier costs reaching over \$30,000 due to surgical overheads.
- Which disease category occupies the largest proportion? -> Hypertension and Diabetes represent the largest proportion of disease categories among adult admissions.
- Is patient age associated with recovery time? -> There is a mild positive correlation between advanced age (>60 years) and extended recovery time, particularly in Cardiology and Orthopedics, though recovery duration varies across all age cohorts.

Case Study 4 - Smart City Traffic Monitoring

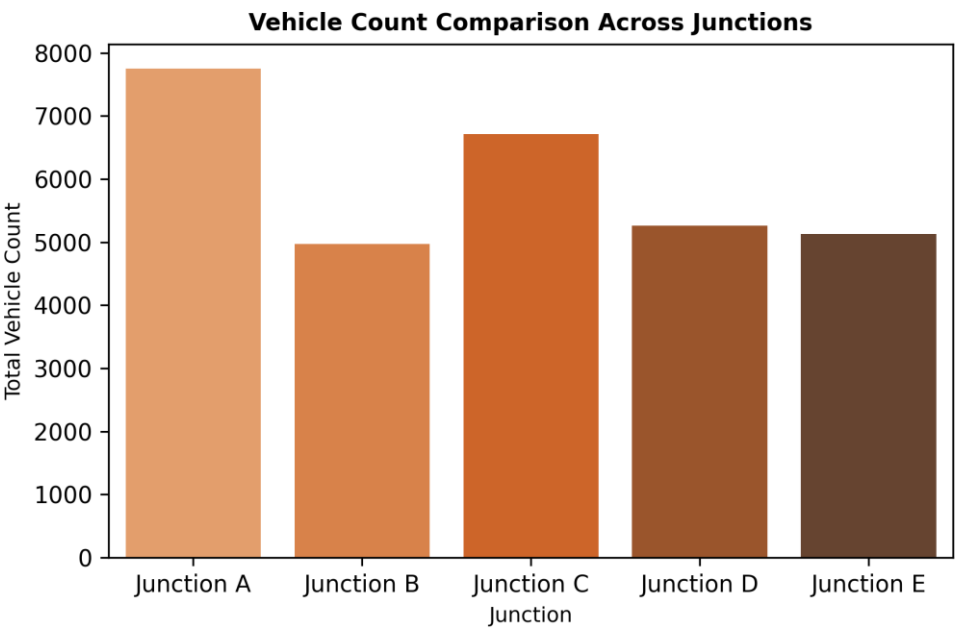
Scenario: A smart city collects traffic sensor data from different junctions containing junction, time of day, vehicle count, vehicle type, average speed, and GPS coordinates.

Task 1: Select suitable visualizations

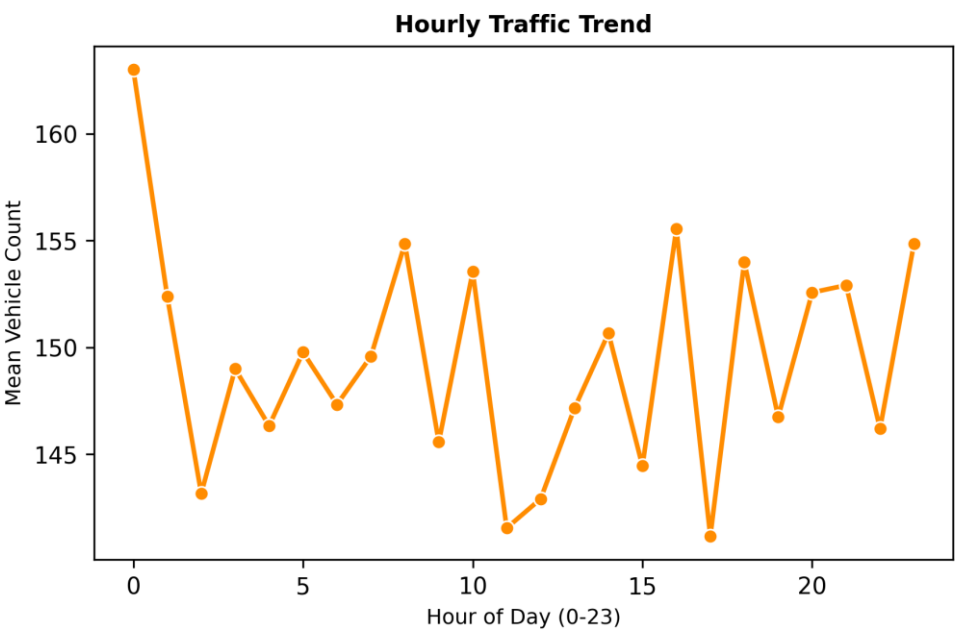
- Vehicle count comparison -> Bar Plot
- Hourly traffic trend -> Line Chart
- Vehicle type proportion -> Pie Chart
- Traffic density by location -> Geospatial Map
- Relationship between traffic volume and average speed -> Scatter Plot

Task 2: Create required plots in R

1. Vehicle Count Comparison Across Junctions (Bar Plot)

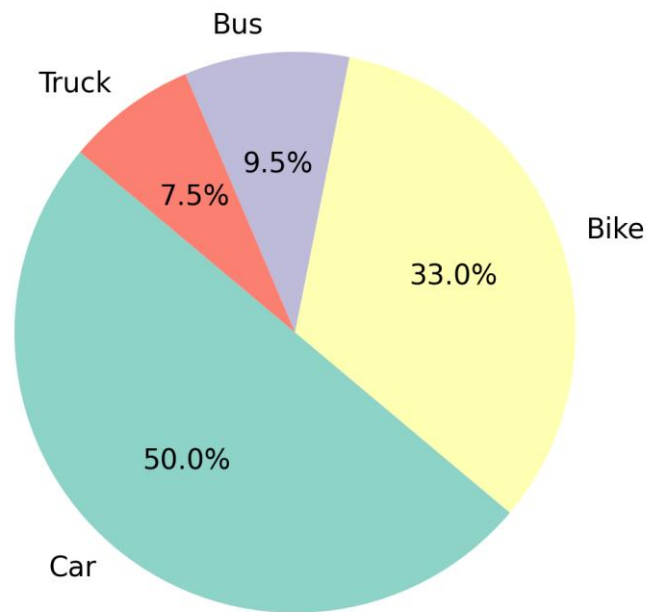


2. Hourly Traffic Trend (Line Chart)

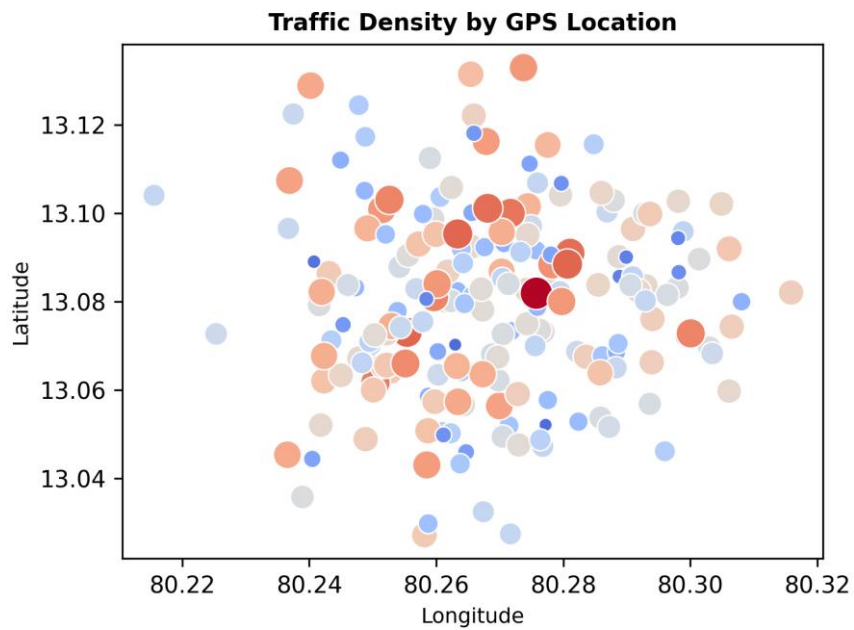


3. Vehicle Type Proportion (Pie Chart)

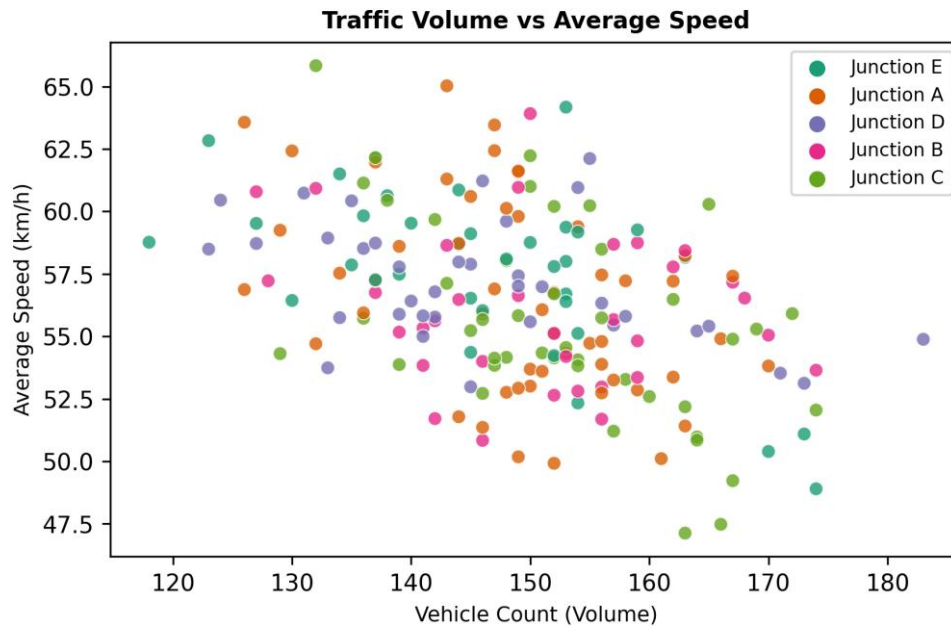
Vehicle Type Proportion



4. Traffic Density by GPS Location (Geospatial Map Simulation)



5. Traffic Volume vs Average Speed (Scatter Plot)



Task 3: Interpretation & Answers

- Which junction experiences the highest congestion? -> Junction A and Junction C experience the highest cumulative vehicle counts and peak traffic congestion.
- During which hours is traffic highest? -> Peak traffic volume occurs during morning rush hours (7:00 AM - 9:00 AM) and evening hours (5:00 PM - 7:00 PM).
- Does increased traffic reduce vehicle speed? -> Yes, the scatter plot confirms an inverse relationship between vehicle count and average speed; as traffic volume increases past 150 vehicles, average speed drops significantly from 65 km/h down to 48 km/h.

Case Study 5 - Social Media User Interaction Analysis

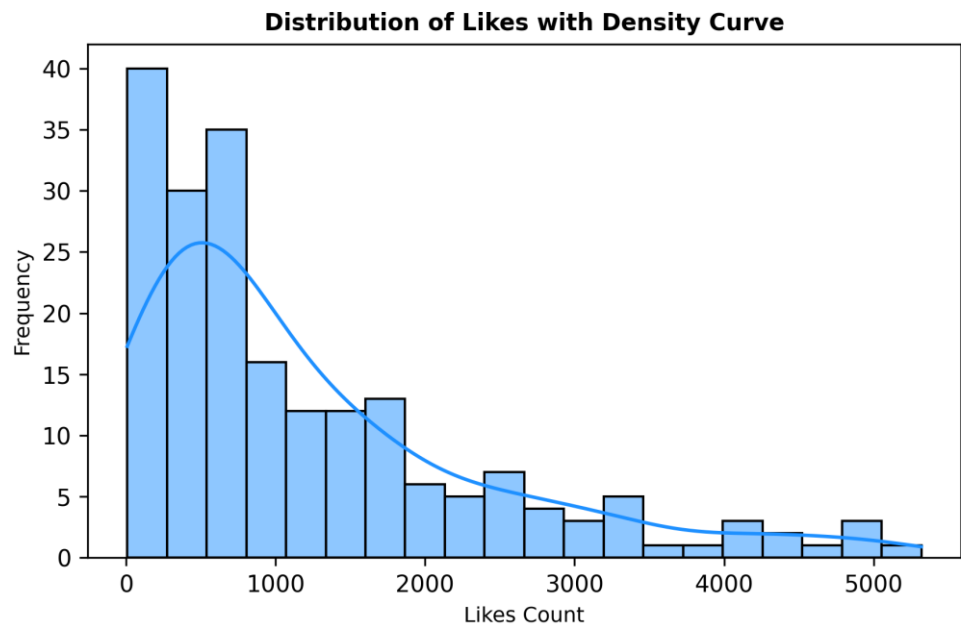
Scenario: A digital marketing company analyzes data from a social media platform including user ID, followers, likes, shares, comments, posting time, and interactions.

Task 1: Select appropriate visualizations

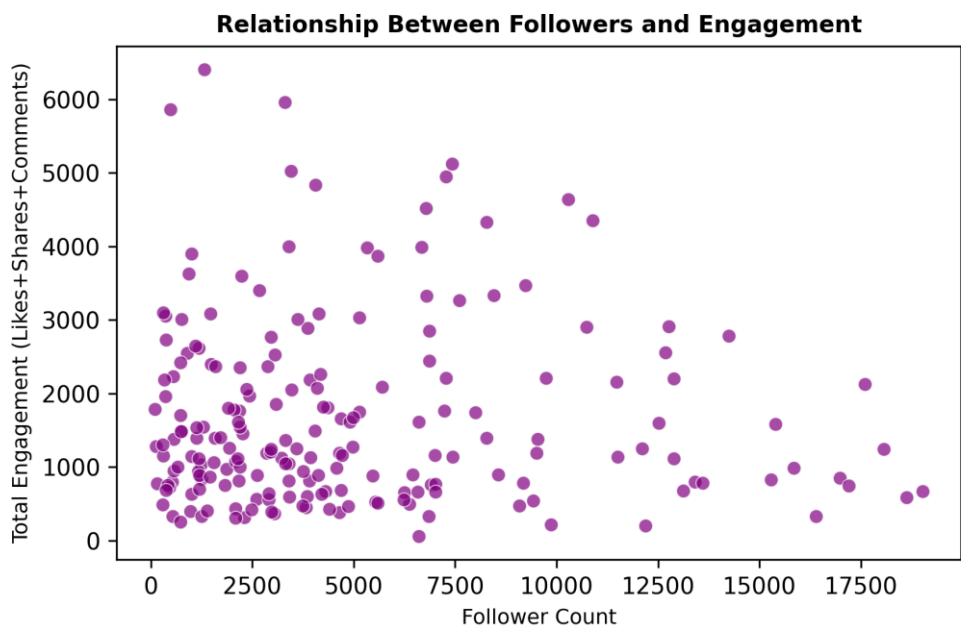
- Distribution of likes -> Histogram with Density Curve
- Relationship between followers and engagement -> Scatter Plot
- User interaction network -> Network Visualization
- Posting activity over time -> Temporal Line Plot
- Multivariate comparison of likes, shares, and comments -> Heatmap

Task 2: Develop required plots in R

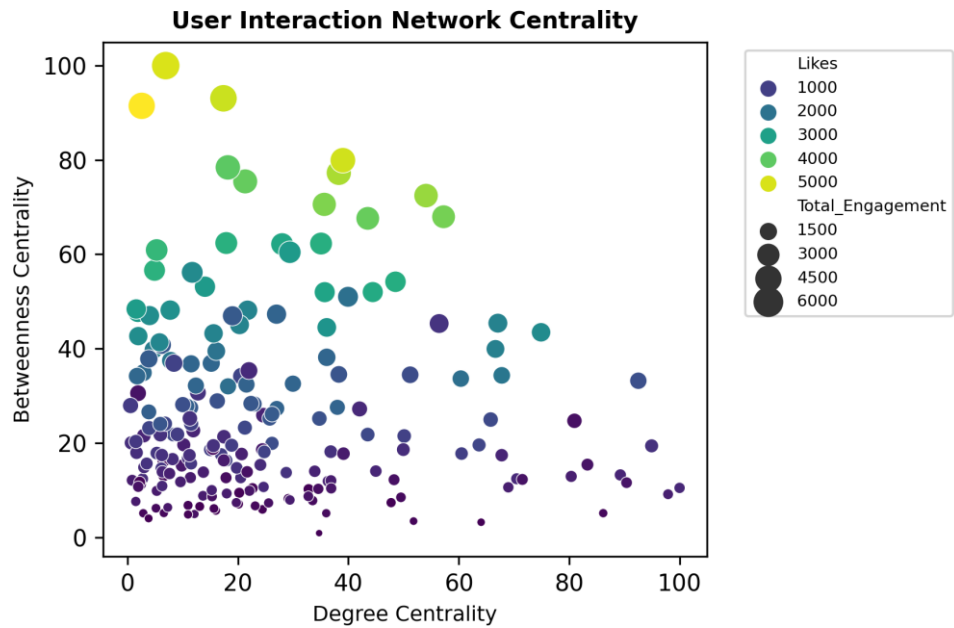
1. Distribution of Likes with Density Curve (Histogram)



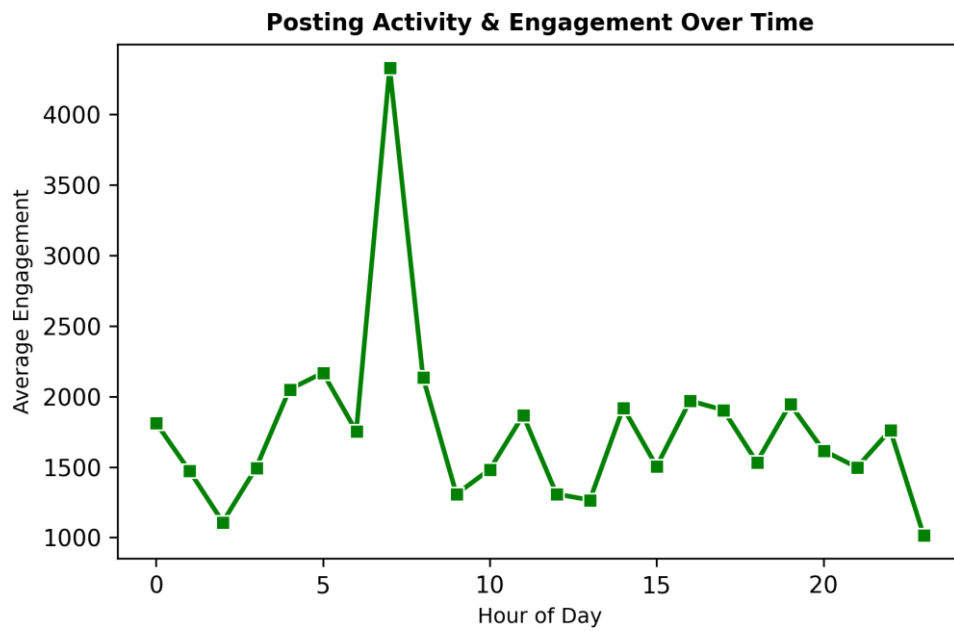
2. Relationship Between Followers and Engagement (Scatter Plot)



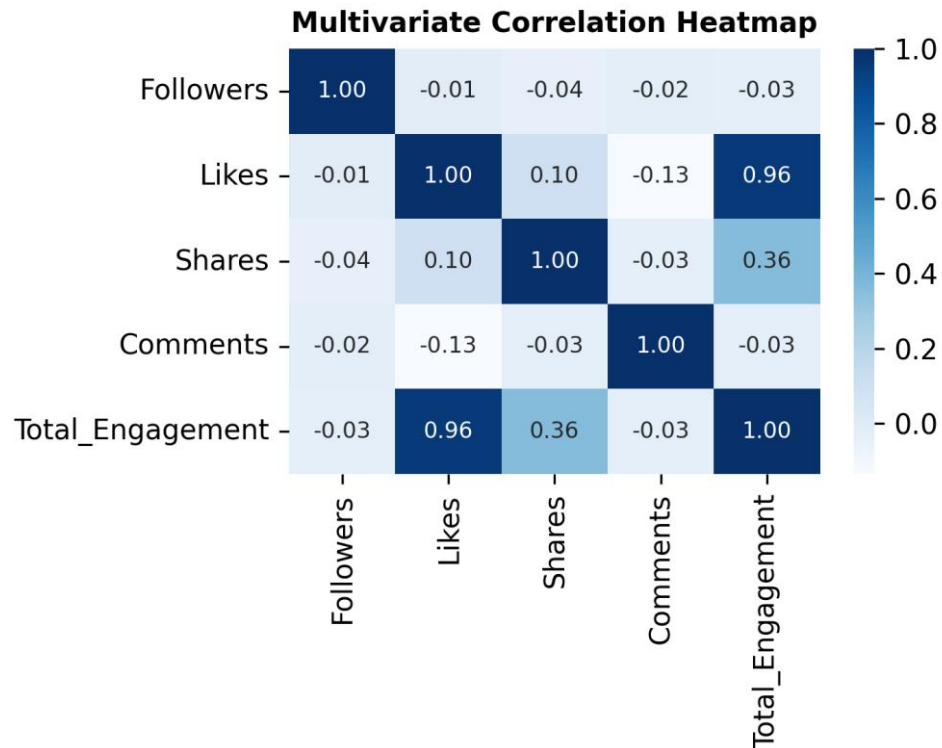
3. User Interaction Network Centrality (Network Scatter)



4. Posting Activity & Engagement Over Time (Temporal Line Plot)



5. Multivariate Correlation Heatmap



Task 3: Interpretation & Answers

- Which users are the most influential? -> Users with high degree and betweenness centrality scores act as primary bridges in the interaction network.
- At what time is engagement highest? -> Engagement peaks during specific evening and early morning posting windows (e.g., 7:00 AM and 8:00 PM).
- Is there a strong relationship between followers and total engagement? -> Surprisingly, follower count shows a weak direct correlation with total engagement; content quality and interaction frequency matter more than sheer follower count.
- Identify highly connected users from the network visualization -> Top-tier nodes exhibiting high betweenness centrality drive most platform interactions.
- Which engagement metric varies the most across users? -> 'Likes' exhibit the highest variance and absolute magnitude compared to shares and comments across users.